Characterization of data-sensitive wireless distributed networked-control-systems

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Abstract- **Distributed networked-controlled-systems (NCS) are a multidisciplinary effort whose aim is to produce a network structure and components that are capable of integrating sensors, actuators, communication, and control algorithms in a manner to suit real-time applications. They have been very popular and widely applied for many years now due to the rapid advancements in data and communication wireless technologies. There are many challenges to be overcome in order to put such a heterogeneous system together. Key issues to be considered are network delay, data sensitivity and information security. This paper characterizes a wireless distributed NCS, a testbed called iSpace based on these key factors. We integrated static network security algorithms DES and 3DES with the NCS testbed iSpace – a multidisciplinary network based robot navigation system – and characterized it on the basis of bandwidth requirement, data classification and data sensitivity, network delay effect on the system performance. The paper demonstrates through results that a dynamic optimization is required between network security for reliability and time-sensitivity of the NCS. Future work in dynamic optimization in security is suggested.**

Index terms – **Distributed networked-control-system, iSpace, data-sensitive wireless network, data security.**

I. INTRODUCTION

A distributed networked-control-system effectively uses many sensors, actuators, computing processors and information technology over a communication network. Connecting the distributed control system components via a network can effectively reduce the complexity of the systems with nominal economical investments making the system scalable with efficient sharing of data. It is easy to fuse the global information and to take intelligent operational decisions over a large physical space. Therefore the research in tele-operation was initiated with the concern of safety and convenience in hazardous environments such as spaces and nuclear reactor plants giving rise to a wide variety of applications like monitoring manufacturing plants, robot navigations, space projects, nursing homes, military applications and many more over the years. This has become possible because of the tremendous increase in the deployments of wireless systems. With the rapid growth, many organizations and individuals are utilizing wireless systems for transmitting sensitive and critical data. As wireless medium is susceptible to easy intercepting, it is extremely critical to protect transmitted data from unauthorized access and modifications in wireless systems. Wireless network security includes essential elements in Internet security devices that provide traffic filtering, integrity, confidentiality, and authentication. Therefore data sharing, data classification and data/network security is of utmost concern in distributed NCS considering the time and data sensitive applications. To characterize a wireless distributed NCS, its properties based on data-sensitivity and network delay effect and the requirement for the network security, we developed a closed-loop network-controlled navigation system called intelligent space (iSpace) as a testbed. It consists of different modules like image processing, automation control, communication and networking for navigation of a differential drive Unmanned ground vehicle (UGV) in an indoor 2-D environment. iSpace is implemented using three basic modules: (i) a vision system to recognize the contents of the environment, the UGV and their configurations [4] (ii) a planning module for computing the optimal, obstacle-free, reference path of the UGV and (iii) a tracking module that uses the sequence of control commands generated by the network controller in the closed-loop navigational system. We illustrate the procedure of the characterization, data classification, data protection and delay compensation for time-sensitive applications through the experiments done on iSpace platforms.

Section II describes the iSpace platform as a distributed NCS and different components involved in it. It also includes description of the wireless system and the encryption algorithms implemented in iSpace. Section III is characterization based on data classification and network delay in a distributed NCS under consideration – iSpace. Section IV is the results section elaborating on the different experiments and their observations and characterization of the NCS. We conclude in and with Section V.

II. SYSTEM DESCRIPTIONAND IMPLEMENTATION

In the wireless systems, several security protocols such as wired equivalent privacy (WEP), 802.1x port access control with extensible authentication protocol (EAP) support are proposed to address security issues [3][2]. Moreover, due to strong security provided by IP security protocol (IPsec) in wired networks, it is considered as a good option for wireless systems as well. Some basic security features associated with these security protocols as defined by IETF for wireless systems are given below [7][1]:

(i) Authentication: The first goal of any security service is to check the identities of mobile users trying to access a network. So that it can be assured that no unauthorized user can access the network. This security mechanism is called authentication of users.

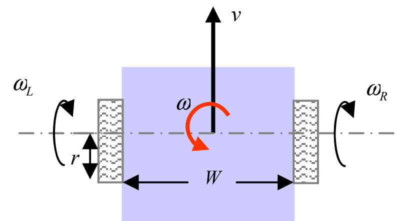
(ii) Confidentiality: Providing data secrecy is another main goal of any security service. The main purpose of confidentiality is to provide protection over data so that no malicious attacker can capture the original information in the transit.
(iii) Integrity: Another main goal of any security service is to provide insurance that no part of transmitted data can be modified in the transit. Even if it is modified, it can be identified by checking the received data. This mechanism is defined as the integrity feature of any security service.
(iv) Non-repudiation: Non-repudiation is insurance provided to a receiver of the data, so that an original sender can not deny the fact that the sender actually sent the data to the receiver.
In this paper, we focus on confidentiality aspect of security service. Other security features such as authentication, integrity and non-repudiation will be discussed in future work.

As described briefly in Section I, a distributed NCS consists of many different components such as sensors to collect data, controllers to process data in the desired manner and actuators to perform tasks asked by the controllers. The key point in building such a system is that the operational intelligence lie on one or many distributed controllers and all these components are connected over a network – these days Internet – to achieve almost limitless virtual space. In this paper, we are using one of the kinds of NCS - Intelligent Space (iSpace) to characterize for confidentiality aspect of security service in distributed NCS. iSpace is a large scale multi-sensor, multi-actuator wireless distributed NCS. We have used iSpace as a network based intelligent control platform for our experiments and simulations for two different security protocols so that we can characterize the system behavior. The more detailed explanation of iSpace therefore follows in this section.

A. iSpace as a testbed

iSpace is a networked-controlled integrated navigation system for a UGV. There are some rules to decide the UGV motion. The UGV has to reach the destination as quickly as possible and avoiding the obstacles in the space. The destination point is chosen by the user via any remote computing interface in the world by accessing the iSpace GUI from the Internet. The iSpace structure is explained in [4]. The components are follows.

(i) An overhead network camera is used as a global sensor. The web camera collects the top-view image of the 2-D space of interest and sends it to the main controller for processing. The network camera used in the iSpace is the Panasonic network camera, KX-HCM270. The camera allows using CGI (common gateway interface) commands in order to control several capture properties under the web environment
(ii) Main network controller with graphical user interface (GUI), which can be on any remote computing interface in the world having the access to internet (Figure 2). The main network controller (Linux Kernel 2.4.18-3 i386) is connected to the global sensors and the actuators in the space of interest through a wireless channel.
(iii) A differential drive UGV (model shown in Figure 1) is the navigator in iSpace. Eq(1) is a principle equation defining the motion of a differential drive UGV. The UGV receives commands from the main controller on the wireless channel to initiate the movement to reach the destination point specified by the user through the GUI. UGV does not have any sensors onboard.



$$\begin{pmatrix} \omega_r \\ \omega_l \end{pmatrix} = \begin{pmatrix} 1/r & W/2r \\ 1/r & -W/2r \end{pmatrix} \begin{pmatrix} v \\ \omega \end{pmatrix} \qquad (1)$$

Figure 1: Model of differential drive UGV. ν - linear velocity, ω - angular velocity, r - radius of the wheel, W -distance between two wheels, ω_r, ω_l - right and left wheel angular velocities respectively.

Figure 2: The graphical user interface for iSpace.

The modules on the main controller are as follows.
(i) A vision system to recognize the contents of the environment including obstacles, the UGV and their configurations. The obstacles and the UGV locations in iSpace are determined by processing the image captured by a wireless IP network camera on top of the iSpace platform. (ii) A planning module for computing the optimal, obstacle-free, reference path for the UGV. (iii) A motion controller - Once the optimal path is generated, the UGV is commanded to follow this path to the destination by choosing reference points on the path at each loop interval Δt.
UGV local controller: One of the major tasks of the local controller residing in the computing processor on the UGV is to receive linear speed (v) and angular speed (ω) information from the main controller via 802.11b wireless channels. Detailed description can be found in [4].

As the application can also be controlled over IP, we use TCP/IP or UDP/IP for the communication between the sensors, actuators and the main controller. TCP is slower than UDP because it also caters for the acknowledgement of the data received. In a time-sensitive application of iSpace, therefore, we send the sensor data with TCP protocol because the sensor data – image in case of iSpace – having all the needed information about the UGV workspace, will decide the course of processing and control commands generation by the main controller. However, the commands from the main controller to the actuator – the UGV in case of iSpace – are sent using the UDP protocol. The motion control commands are time sensitive. Once the interval time (Δt) is passed the old control command might have become obsolete and the new position of the UGV needs new motion commands to keep the UGV on path. It is one way communication. If the packet is dropped, the UGV will simply receive the next command packet corresponding to the next current position and act accordingly, all packets carrying the time sensitive information about the real-time motion of the UGV.

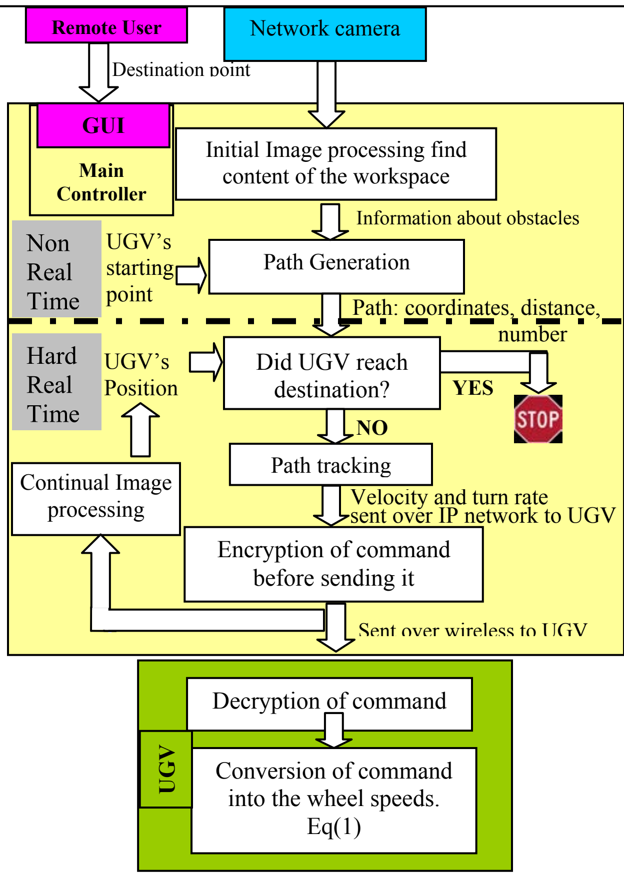


Figure 3: The data flow for i-Space- the distributed NCS used as a testbed.

B. Data Encryption

Encryption algorithms for wireless systems can be broadly classified into two categories: symmetric and asymmetric ciphers. Encryption algorithms, which require a single cryptographic key to encrypt and decrypt the data, are termed as symmetric ciphers. The Data Encryption standard (DES) and the Advanced Encryption Standard (AES) are two symmetric ciphers which are considered secure for wireless systems [2][5]. However, the issue associated with symmetric ciphers is how to transmit cryptographic keys securely to the legitimate recipient. On the other hand, asymmetric ciphers are the ones which require different cryptographic keys for encryption and decryption of data. For example, public key encryptions, such as RSA, employ asymmetric ciphers. In public key encryption, each user has a pair of public and private keys. The public key is published while the private key is kept confidential. Messages are encrypted using the public key of the recipient, and can be decrypted only by the private key of the recipient. As private key is never known to others, this method eliminates the problem of transmitting secure keys to receivers. However, the critical issue associated with public key encryption is the high computational power required for encryption and decryption [1]. Due to this, public key encryption is used only for exchanging cryptographic keys securely in real time systems, whereas symmetric ciphers are used for actual data encryption and decryption.

Therefore, in this work also, we implement DES and 3DES (a stronger variant of DES) for encrypting and decrypting the data securely over NCS. We do not consider AES in this work due to two reasons. First, AES requires wireless cards with new software drivers, which are not compatible with system software in UGVs. Moreover, due to limited processing and memory capabilities of UGVs in NCS, it was not feasible to have more number of security algorithms implemented in robotics node at the same time.

As discussed above, we use DES and 3DES encryption algorithms to implement confidentiality feature over NCS. Data Encryption Standard (DES) incorporates a 64-bit key to encrypt and decrypt the transmitted data. However, as the least significant bit of each byte in the key is a parity bit, that bit from each byte is ignored during actual encryption and decryption process, leading to reduction of 8 bits from 64-bit key. Therefore, DES uses 56-bit key effectively [2]. Besides, DES, being a block cipher, operates on 64-bit blocks of plaintext as input and outputs 64-bit blocks of encrypted blocks. 3DES is very similar to DES and requires three 64-bit keys to provide stronger security than DES. In 3DES, first each 64-bit block is encrypted by using first 64-bit key, then decrypted using second 64-bit key and then encrypted again by using third 64-bit key. For decryption at the receiver side, the entire process is reversed to obtain the original form. Consequently, 3DES is 3 times slower than DES [5][7]. DES incorporates different modes of operations such as ECB (Electronic Code Book), CBC (Cipher Block Chaining), CFB (Cipher Feedback) and OFB (Output Feedback). As ECB is considered the fastest mode of operation, it is used commonly in real time systems [7]. Therefore, in this work, we use DES and 3DES with ECB mode to reduce computational processing.

The whole control algorithm runs in a loop continuously until the UGV reaches its destination chosen by the user. This is explained by the flow diagram (Figure 3).

III. Data Classification

For a distributed NCS, the data classification will mainly be categorized as follows.

(i) *Sensor data*: The data from different sensors to the main controller. This data contains information about the space of interest and information required to take intelligent decisions.

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In iSpace, the top view image of the workspace sent by the network camera to the main controller is the sensor data. The jpeg image is of the size 320 X 240. This is the bulkiest data handled in iSpace. The image contains all the required information about the workspace required for navigation.

(2) *Command data*: The data from the main controller towards various actuators to carry the commands for the actuators to perform the operations intelligently. As the UGV completely depends upon the controller's commands to actuate the motion. The controller's data packet to the UGV contains speed (ν) and turn-rate (ω) for the UGV. The command string is of the maximum size of 12 bytes.

In a distributed NCS, data can also be classified based on its priority, importance or bandwidth requirement.

Bandwidth requirement: The jpeg image of the iSpace workspace (320 X 240) (sensor data) requires high bandwidth compared to that the command data with 12 bytes in each loop.

Time sensitivity: The continuous image data contains information about the UGV motion which is time sensitive, unlike the initial image acquisition which is processed to know about the static components of the workspace before the motion starts. The command data is also time sensitive because it controls the motion of the UGV in real-time. Therefore data validity expires after Δt – loop time for each iteration during the motion.

We also illustrate the different delays which have their effects on the functionality of the whole system. On a broader level, the delay classification can be done as Transportation or Network delay component and Computational or Processing delay component. Their definitions and significance for a distributed NCS is explained as below.

(i) Transportation / Network delay component ($\tau_k^{NetworkDelay}$)

a. Sensor to controller delay (τ_k^{sc}) - The time between the main controller requests the camera for the image (Image acquisition delay ($\tau_k^{imageacq}$) and it is captured and saved on the memory of the controller. Therefore we can say $\tau_k^{sc} = \tau_k^{imageacq} + n \cdot m\left(\tau_k^{RTT}\right)$, where τ_k^{RTT} is the round trip network time and m is the number of packets needed to send the complete image information. n is the number of cameras to access to get the whole workspace image.

b. Controller to actuator delay (τ_k^{ca}) – This is the time for the reference command takes to reach from the main controller to the UGV $\tau_k^{ca} = \tau_k^{RTT}/2$.

(ii) Computational / Processing delay component ($\tau_k^{ComputationalDelay}$).

a. Non-real time Computational delay (τ_k^{NRc}) – It consists of the initial image processing time, path generation and the system processing time. $\tau_k^{NRc} = \tau_k^{imeprocess} + \tau_k^{pathgen} + \tau_k^{sysprocess}$.

b. Real time Computational delay (τ_k^{Rc}) – This is the time for the main controller to execute the control algorithm. i.e. the consecutive iterations of image processing and path tracking algorithm. So if the loop runs for l times before the UGV reaches the destination. $\tau_k^{Rc} = l \cdot \left(\tau_k^{imeprocess\_r} + \tau_k^{pathtrack} + \tau_k^{sysprocess\_r}\right)$.

(iii) Confidentiality delay: Applying network security such as confidentiality will add to the delay and it can be categorized as encryption delay and decryption delay. Encryption delay ($\tau_k^{Encrypt}$) is the total time required to encrypt the data from its plain form, and the decryption delay ($\tau_k^{Decrypt}$) is the total time required to covert the encrypted data to its original form.

Moreover, stability and reliability are the two main factors for a distributed NCS. When a UGV is controlled over a network, its performance can be degraded by network-induced delays, and the system can even become unstable. Therefore the network delay compensation is necessary in a remote controlled, network based time sensitive application like iSpace. Similarly, because of the lack of security, the data can be maliciously used affecting the reliability of the operation. These two factors considered simultaneously will have different effects on the system described in the next section.

IV. Experimental results and Discussions

The iSpace workspace described in this paper is a 3m x 4m area. In this paper, we demonstrate the features of the suggested NCS using three case studies - first without any encryption in loop and later with DES and 3DES encryption algorithms respectively. The distance between the UGV's position and the reference path generated by the main controller is calculated at each Δt. This difference is called the 'tracking error'. The total time to reach the destination is also recorded. The 'tracking error' and the 'total time' are the main criteria used for performance evaluation in this paper.

Following are the delays observed over multiple experiment runs on iSpace:

Average image acquisition delay ≈ 0.2307 seconds.

Average non-real time image processing delay (initial image processing) ≈ 2.06 seconds.

Average real-time image processing ≈ 0.02 seconds.

We observe that the image acquisition delay is almost 90 – 92 % of the whole feedback loop delay. Image acquisition delay is hardware dependent i.e. the type of the network camera used in the iSpace system. This delay can be reduced by using a network camera with faster response.

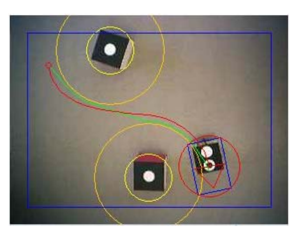
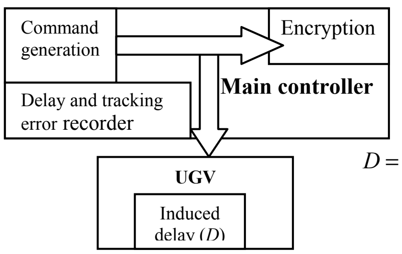


Figure 4: Experiment setup of the obstacles and the UGV

For comparison purpose, we run the experiment for a fixed pair of starting and end point for the UGV and fixed location of the obstacles for all the 3 cases. Setup is displayed in Figure 4. For DES and 3DES, the implementation should involve encryption on the main Linux controller and decryption on the UGV. Currently the microcontroller on the UGV does not support decryption algorithm therefore unencrypted command data was sent to the UGV requiring no decryption to be implemented. This will be implemented in future investigations. As the results presented in this paper mainly emphasize on characterizing the system performance due to the delay effect, we induced an artificial delay of fixed amount D in the controller loop simulating the decryption delay which would otherwise occur on the UGV side (Figure 5). D was calculated using Eq(2).



$$D = D_{avg}\frac{f_{clk}(controller)}{f_{clk}(UGV)} \quad (2)$$

Figure 5: Block diagram of the experimental setup for introducing security delay.

$f_{clk}(controller)$ = clock frequency of the Linux controller.
$f_{clk}(UGV)$ =clock frequency of the microcontroller on UGV.
D_{avg} = the average time recorded for execution of the decryption algorithm on the Linux controller for the command data type used in iSpace.
The ratio of clock frequencies of the Linux controller and the UGV microcontroller is the only factor considered to calculate the artificial delay equivalent to the decryption delay on the UGV. We assume that other memory and operating system overheads are negligible on both Linux and the UGV controller and the processing capabilities of the microcontroller are also equivalent to the Linux controller.

Table 1 records total time, tracking error and loop delay for all the 3 cases. The system without security exhibits less tracking error and takes less time to track the same path than DES and 3DES. Though DES displays comparable results to a non-security implementation, 3DES is approximately 2.7263 = ((0.2725-0.2207) / (0.2397-0.2207)) times slower than DES (Table 1). Subtraction the average loop delays for 2 cases will give us the delays which occur due to the difference between security implementation of the 2 cases, cancelling out the common delays like network delay and image processing delay.
In Figure 6, we plot the difference of loop delay and image processing delay as a function of time. We subtract image processing delay because it is independent of the security algorithm implementation. Therefore, Figure 6 displays the effect of only the network delay and the delay introduced by the security algorithms on the system.

Table 1: Observations for time and tracking error for one experiment run

	Total time (s)	mean tracking error (m)	Average loop delay (s)
No security	14.7877	0.0310	0.2207
DES	15.2138	0.0370	0.2397
3DES	17.7099	0.0448	0.2725

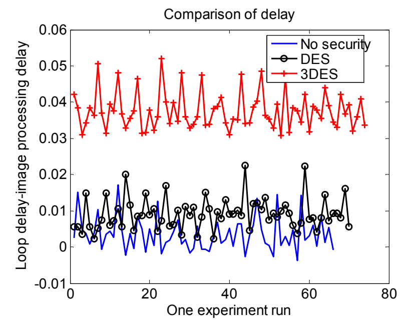


Figure 6: Loop delay-image delay plotted for one experiment run

To validate further, we run the same experiment for each of the 3 cases multiple times. Figure 7 plots minimum, mean and maximum values of tracking error and total time respectively over the multiple experiment runs for 3 algorithms. It shows that the encryption algorithm in NCS will slow down the system response, as expected.

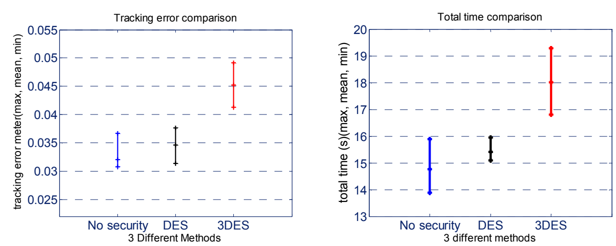


Figure 7: (a) Path tracking error comparison (b) Total time comparison over 5 repeated experiment runs

V. Conclusion and Future Work

Through the experiments and simulations on the iSpace platform – which demonstrates a distributed networked-controlled-system – we characterized a data-sensitive wireless distributed NCS, discussed its properties based on data-sensitivity, bandwidth requirements, network delay effect and the requirement for the network security. Though the experiments are presented only for two security algorithms, and it is obvious that it is utmost important to have network security on the wireless data medium to protect the information in an NCS; this paper puts forth a point that statically adding security algorithms in a time sensitive NCS would degrade the performance of the system to some extent, which is critical considering the stability of the NCS. This trade-off between the security addition and real-time performance of the system can be solved by dynamically optimizing the part of the security in the system loop. This will further need to quantify the security-induced delay and system overhead and its impact on the overall NCS under different operating conditions such as UGV

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speed, curvature of the path, gain of the system, number of sensors and actuators etc.

Thus, future work will involve investigating the dynamic data protection method in addition to finding an optimum solution between the reliability offered by network security, stability and time-sensitivity required for an NCS. Dynamically choosing the data security algorithms will be a function of data priority, bandwidth requirement and time sensitivity of the data in an NCS. Therefore the characterization of any data-sensitive wireless NCS over different parameters is a key.

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